brief contents

PART 1 BUILDING BASIC GENERATIVE AI WEB APPS 1

1 ■ Using generative AI in web apps 3
2 ■ Building your first generative AI web application 19
3 ■ Connecting AI models with the Vercel AI SDK 54
4 ■ Managing conversation and state in your application 88

PART 2 ADVANCED GENERATIVE AI TECHNIQUES AND DEPLOYMENT 127

5 ■ Prompt engineering in web applications 129
6 ■ Building AI workflows with LangChain.js 166
7 ■ Document summarization and RAG with LangChain.js 197
8 ■ Testing and debugging techniques 232
9 ■ Deployment and security 268

PART 3 HANDS-ON PROJECTS 303

10 ■ Building an AI interview assistant: Project walk-through 305
11 ■ Building an AI RAG agent: Project walk-through 317

PART 4 ADVANCED INTEGRATIONS AND THE FUTURE OF AI 329

12 ■ Integrating web apps with the Model Context Protocol 331
appendix ■ Running the examples 353